1. Install Docker

On Windows or macOS:

- **Download Docker Desktop:** https://www.docker.com/products/docker-desktop/
- After installation, open Docker Desktop and ensure it's running.

On Linux (Ubuntu example):

```bash
sudo apt update

sudo apt install -y docker.io

sudo systemctl start docker

sudo systemctl enable docker
```

Optional: run Docker without sudo

```
sudo usermod -aG docker $USER
```

You may need to log out and back in for user group changes to take effect.

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2. Create Dockerfile to Containerize an Application

Let's containerize a simple **Python Flask app** as an example.

Project Structure:

myapp/

```
├── app.py
├── requirements.txt
└── Dockerfile
```

---

app.py (Flask App)

python

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello from Dockerized Flask App!"
```

---

requirements.txt

ini

```ini
Flask==2.3.2
```

Dockerfile

```dockerfile
# Use official Python image
FROM python:3.10-slim

# Set working directory
```

WORKDIR /app

Copy files

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

Expose port and run

EXPOSE 5000

CMD ["python", "app.py"]

---

3. Build & Run Docker Container

bash

Navigate to project directory

cd myapp

Build Docker image

docker build -t my-flask-app .

Run the container

docker run -p 5000:5000 my-flask-app

Visit http://localhost:5000 — you should see:
"Hello from Dockerized Flask App!"

Summary

Step	What You Did
Install Docker	Downloaded Docker Desktop or installed CLI
Write Dockerfile	Described how to build your app's image
Build Image	Used docker build
Run Container	Used docker run